WE-MAKE-DEVS · 'FLIP THE SWITCH' HACKATHON · TRACK 01: BEST BUSINESS WORKFLOW

# Audit.CO2

## Autonomous Multi-Agent Enterprise Carbon Accounting with Cryptographic Proof Boundary

Powered by Switch AI · 4 Specialized Agents · Human-in-the-Loop Collaboration

### 4 Specialized Agents

Orchestrator (@carbon-room), Collector, Watchdog, and Reporter running 4 concurrent duplex WebSockets inside a shared enterprise room.

### Zero Hallucinations

Models are strictly quarantined to extraction; a deterministic proof validator guarantees verbatim raw source text matching and high-precision math.

### 91-Event Audit Chain

Tamper-evident append-only SQLite ledger. Every human resolution, approval, and event is cryptographically hashed with SHA-256.

ACT I: PROBLEM & MOTIVATION

# Enterprise Carbon Accounting Is Broken

CSRD, SEC, and GHG Protocol mandates require audit-ready disclosures, but current workflows fail.

## ■■ The Status Quo: Spreadsheet Chaos

- **3+ Weeks per Monthly Close:** Sustainability teams manually chase utility bills, PDF receipts, and cloud invoices across disparate teams.
- **Spreadsheet Fragility:** Broken Excel formulas, manual copy-pasting, and missing emission factor version tracking.
- **Zero Chain of Custody:** Auditors cannot verify who approved an estimate, when, or against what source evidence.
- **Severe Regulatory Risk:** CSRD and SEC penalties for inaccurate ESG disclosures reach millions of dollars in fines.

## ■ Why Solo AI Chatbots Fail

- **LLM Number Hallucinations:** Generative AI invents numbers and dates when reading complex utility bills.
- **Floating Point Arithmetic Drift:** LLMs fail to perform exact 10-decimal GHG Protocol factor multiplications.
- **Solo Chatbot Isolation:** A solo bot in a web tab cannot coordinate facility managers, accountants, and auditors.
- **Black-Box Liability:** External assurance partners reject unverified AI answers as legal compliance evidence.

ACT II: THE SOLUTION & WHY SWITCH

# Audit.CO2: Collaborative Multi-Agent Enterprise Room

Instead of a solo chatbot, 4 specialized agents collaborate with human stakeholders in Switch.

## ■ @carbon-room

**ORCHESTRATOR**

- Manages close phases (COLLECTING → PUBLISHED)
- Locks factor set editions
- Live Close Board maintainer
- Enforces 12-post noise limit

## ■ @collector

**INGESTION ENGINE**

- Proof Boundary verification
- SHA-256 content deduplication
- Enforces verbatim raw text match
- Parses PDFs, EMLs, CSVs, cloud JSON

## ■■ @watchdog

**AUDIT & RULES**

- 7 Deterministic GHG rules
- Flags period mismatches & variances
- Opens targeted gap threads
- Recalculates ledger in real time

## ■ @reporter

**DISCLOSURE & TRACE**

- Mandatory compliance disclosures
- Unverified AI datacenter flags
- Cryptographic 91-event audit check
- Publishes PDF, JSON, HTML, Markdown

ACT III: LIVE DEMO & HITL MAGIC

# The October 2026 Close: Human-in-the-Loop Collaboration

How agents and humans resolve compliance gaps live in Mattermost / Switch.

### Step 1: Start Close

**@carbon-room close October**

Orchestrator locks factor set (FS-2026-10), transitions state to COLLECTING, and signals Collector.

### Step 2: 4 Gaps Flagged

**@watchdog opens threads**

Collector harvests invoices. Watchdog catches 4 traps (G-1..G-4) and refuses clean total: 0.0 tCO2e verified.

### Step 3: Human Resolutions

**upload & pick & approve**

Priya uploads real bill; Lead picks Rev A, approves factor & variance. Total moves live to 21.9 tCO2e (100% Coverage).

### Step 4: Draft & Publish

**@reporter draft → publish**

Mandatory regulatory disclosures attached. Lead publishes immutable reports (PDF, JSON, HTML, MD).

ACT IV: DEEP TECHNICAL ARCHITECTURE

# Zero-Hallucination Proof Boundary & Deterministic Core

Why Audit.CO2 is production-grade and legally verifiable.

## Proof Boundary

**ZERO HALLUCINATIONS**

AI models strictly extract candidates. A deterministic validator checks that every number matches verbatim in raw file bytes:

excerpt in norm(page)

If model hallucinates, validate() rejects.

## Deterministic Math

**DECIMAL PRECISION**

Zero floating-point rounding drift.

All multiplications executed using Python Decimal:
Activity × PUE × Factor

Verified down to 10 decimal places (21.9418970878 tCO2e).

## Cryptographic Ledger

**SHA-256 EVENT CHAIN**

Every human resolution, upload, and approval is chained in SQLite:

H_n = SHA256(H_n-1 || ev)

Any manual DB tampering breaks the cryptographic chain (91 events verified).

## Switch Bus Engine

**4 DUPLEX WEBSOCKETS**

Maintains 4 concurrent duplex Switch sockets.

Implements SPEC-09B cross-agent addressing delegation.

Pure intent grammar parser backed by 642 passing unit & integration tests.

ACT V: IMPACT & CONCLUSION

# Turning 3 Weeks of Spreadsheet Chaos into 3 Minutes

The future of enterprise sustainability reporting powered by Switch.

## 95% Time Reduction

**3 WEEKS → 3 MINUTES**

Automates ingestion, rule checking, and variance analysis while keeping humans in the loop for binding sign-offs and legal authority.

## 100% Audit Verifiable

**MATHEMATICALLY REPRODUCIBLE**

Every published gram of CO2 can be traced back to raw source bytes via @reporter trace S-001. No black-box estimations.

## Complete Output Suite

**4 REGULATORY FORMATS**

Outputs out/2026-10/v1/ with report.pdf, report.json, report.md, and interactive coverage.html with compliance disclosures.

## Built on Switch

**HUMAN-AGENT SYNERGY**

Proves how autonomous agents and human experts can collaborate inside enterprise chat platforms safely and effectively.